ML 2025 Project

Authors: Leonardo Celati · Damiano Degliotti · Daniele Melaccio

Team Name: CDM25

Master Degree: Informatica Umanistica

Type of Project: B

Objectives

- **Common**
  - Evaluate state-of-the-art machine learning frameworks and use them to develop a structured collection of notebooks addressing the MONK and CUP type B assignment requirements.
    - PyTorch was used for neural-network implementation, while scikit-learn was adopted for other machine learning models (SVM, KNN,..).
- **Monk**
  - Develop a reusable framework for model selection and comparison across KNN, kernel-based SVMs and neural networks on the three MONK classification tasks. For MONK-3, provide both regularized and unregularized neural-network configurations
  - The resulting framework serves as the methodological and software foundation for the CUP task.
- **CUP**
  - Reuse the shared framework to build a regression pipeline for ML-CUP25, using Mean Euclidean Error as the primary model-selection metric.
  - Combine complementary models through ensemble weights learned exclusively from out-of-fold predictions.
  - Produce a reproducible blind-test submission in the required CUP format.

Model selection — methodological overview

- **Data preparation**
  - MONK categorical attributes were transformed using canonical one-hot encoding.
  - CUP preprocessing was performed within each training fold to prevent data leakage.
  - The CUP labelled set was split once into 400 development and 100 internal-test cases.
- **Hyperparameter selection**
  - Scikit-learn models: exhaustive GridSearchCV over discrete, model-specific search spaces.
  - Neural networks: sequential Optuna optimization to keep the search computationally feasible.
  - The task-specific validation metric (MEE for the CUP and MSE for MONK) was the primary selection criterion; training–validation gaps and complementary metrics such as MSE, RMSE and R^2 were used diagnostically for the CUP.
- **Verification and assessment**
  - Nested CV was used on CUP to quantify selection optimism for scikit-learn models.
  - Dedicated manual NN notebooks were built to verify the Optuna-selected configurations.
  - All choices were frozen before the single evaluation on the untouched internal test.

Model selection — GridSearchCV methodology

GridSearchCV was used for the scikit-learn models because their hyperparameter spaces were discrete, bounded and computationally manageable for it evaluates every combination in each predefined model-specific grid. All preprocessing operations were included in a Pipeline, ensuring that scaling was fitted independently within each training fold and preventing information leakage. The workflow was:

i. **Model-specific search spaces:** separate grids were defined for each models according to the parameters relevant to each estimator.
ii. **Fold-local preprocessing:** scaling and model fitting were combined in a single pipeline, so each candidate configuration was evaluated using only the training portion of each fold.
iii. **Exhaustive cross-validated search:** every hyperparameter combination was evaluated on the development set using the same shared cross-validation strategy and MEE as the primary selection criterion.
iv. **Nested-CV diagnostic assessment:** an inner three-fold GridSearchCV was embedded within the outer cross-validation loop to estimate selection bias and compare nested and non-nested results.
v. **Configuration freezing and refit:** eventually the selected configuration was frozen and refitted on the complete development set to perofrm a single internal-test assessment.

Model selection — Optuna methodology

Optuna hyperparameters were optimized sequentially with a coarse-fine approach to reduce the dimensionality of each search and make the computational cost manageable. For CUP we further compared three different run each with a fixed PyTorch TRAINING_SEED to provides a reproducible and computationally feasible model-selection procedure. Each selected config was then tested on manual notebook with three different seed resulting in 9 search in total from which we picked the best configuration based on metrics evaluation (see Monk and Cup section). Pipeline sequence:

i. **Learning rate:** establishes stable and effective convergence and the reliability of all subsequent comparisons. The search interval has been tuned by observing lr that didn’t fall at the boundary.
ii. **Batch size:** controls gradient noise and optimization dynamics.
iii. **Weight decay:** tuned under a reasonably stable training configuration, allowing its effect on generalization to be evaluated more clearly. Again the interval allow the results not to fall at the boundary.
iv. **Architecture and activation:** they were explored at this stage because these choices are computationally more expensive and their comparison is meaningful only after the main optimization and regularization parameters have been stabilized.
v. **Final cross-validation:** verifies the complete selected configuration before internal assessment.

MONK KNN results and hyperparameters overview

KNN models were selected by 5-fold stratified cross-validation over k, distance weighting and Minkowski order, then refitted on each complete training set. Note: With distance-weighted KNN, the training error might be zero by construction, as each training point dominates its own prediction.

Task	Architecture and hyperparameters	MSE (TR / TS)	Accuracy (TR / TS)
MONK-1	k=7; Manhattan (p=1); distance weights	0.00000 / 0.17130	100.00% / 82.87%
MONK-2	k=21; Manhattan (p=1); uniform weights	0.32544 / 0.34954	67.46% / 65.05%
MONK-3	k=35; Manhattan (p=1); distance weights	0.00000 / 0.06019	100.00% / 93.98%

MONK SVM results and hyperparameters overview

SVM models were selected by 5-fold stratified cross-validation over kernel-specific parameters and regularization strength C, then refitted on each complete training set. Note: all the Monk Learners lead towards Polynomial Kernel as the best performers.

Task	Architecture and hyperparameters	MSE (TR / TS)	Accuracy (TR / TS)
MONK-1	Polynomial; C=1; gamma=1; degree=2	0.00000 / 0.00000	100.00% / 100.00%
MONK-2	Polynomial; C=10; gamma=1; degree=2	0.00000 / 0.00000	100.00% / 100.00%
MONK-3	Polynomial; C=10; gamma=0.1; degree=3	0.04098 / 0.01852	95.90% / 98.15%

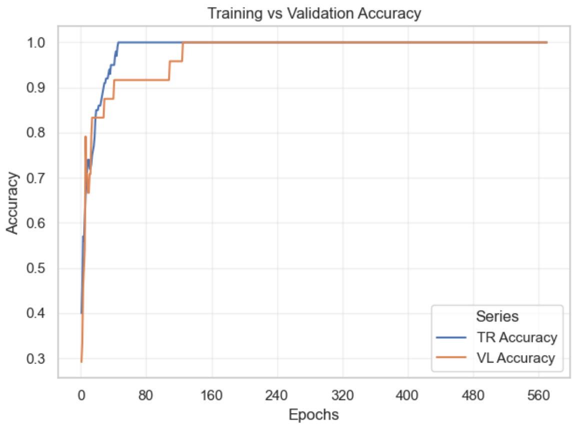
MONK NN results and hyperparameters overview

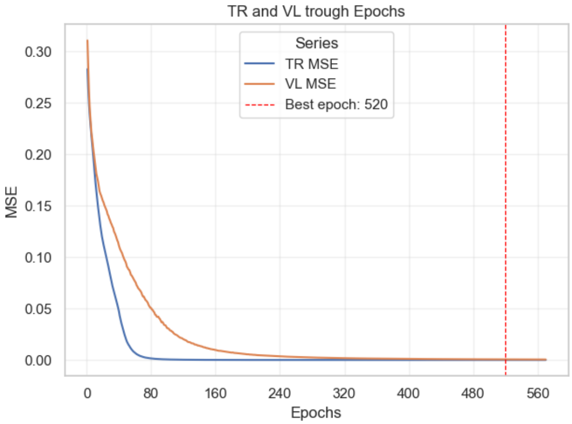
Optuna-derived NNs use SGD with common momentum and Nesterov acceleration, BCEWithLogits loss and mini-batches of 20. Note for MONK-3: (1) the unregularized run is a diagnostic comparison (2) Optuna was not particularly efficient in exploring the search space due to small train set so we ended up manually experimenting hyperparameters tuning.

Task	Architecture and hyperparameters	MSE (TR / TS)	Accuracy (TR / TS)
MONK-1	8 ReLU; η=0.02991; λ=6.42×10^{-6}; μ = 0.9, 458 ep.	2.52×10^{-6} / 4.76×10^{-4}	100.00% / 100.00%
MONK-2	8–16 ReLU; η=0.02976; λ=2.93×10^{-5}; μ = 0.9, 112 ep.	7.09×10^{-7} / 1.12×10^{-6}	100.00% / 100.00%
MONK-3	5 ReLU; η=0.02818; λ=0.05; μ = 0.9, 23 ep.	0.07232 / 0.06169	93.44% / 97.22%
MONK-3 no reg.	5 ReLU; η=0.02818; λ=0; μ = 0.9, 1000 ep.; no ES	0.04934 / 0.02324	97.54% / 94.44%

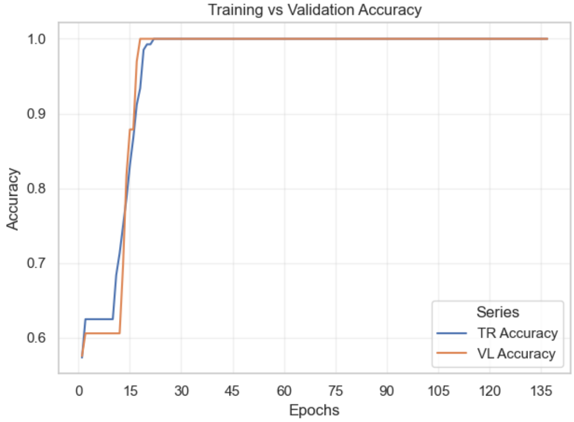
MONK NN results (1/2)

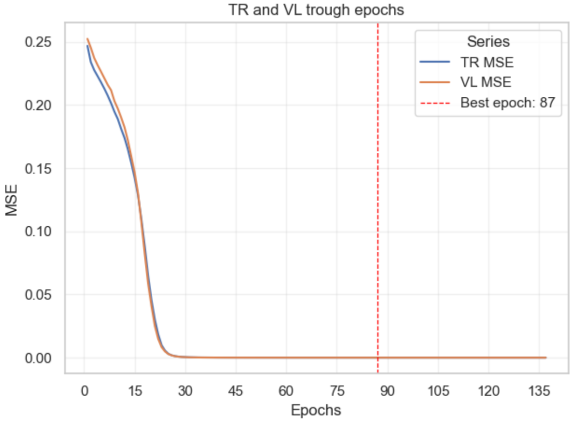
MONK-1 100% training and validation reaches accuracy, with both MSE curves converging smoothly. The hold-out curve selects epoch 520; the final model is refitted for 458 CV-median epochs, the median selected across CV folds.





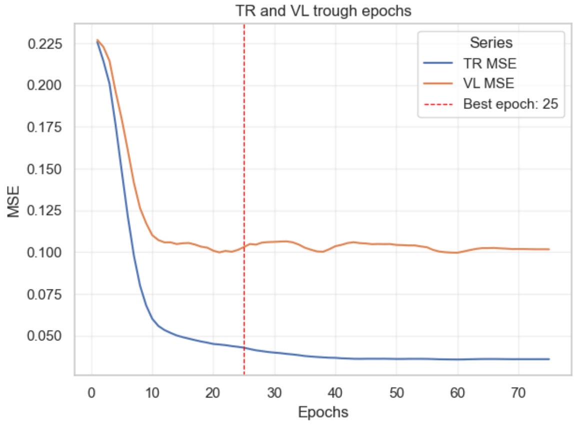
MONK-2 rapidly reaches 100% training and validation accuracy, while the closely aligned MSE curves converge near zero. The hold-out curve selects epoch 87; the final refit uses 112 CV-median epochs.



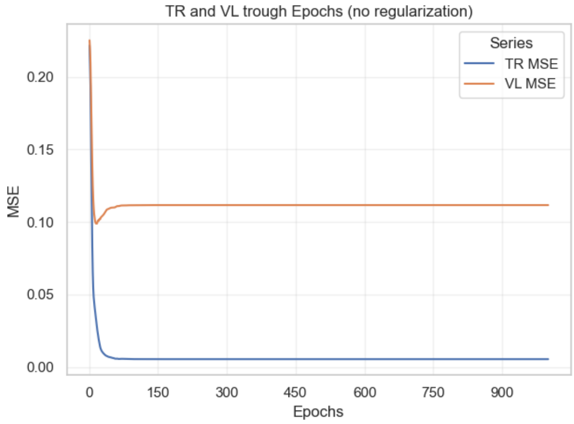


MONK NN results (2/2)

MONK-3 Regularization controls overfitting: hold-out validation MSE is minimized at epoch 25 where a model snapshot is taken. The final refit reaches 97% test accuracy.



MONK-3 Without early stopping and weight decay, training MSE keeps falling while validation MSE rises with a consistent gap trough remaining epochs. The final refit reach 94% test accuracy.



CUP validation leakage controls

The following controls preserve the independence of model selection, internal assessment and blind-test prediction on CUP assignment:

- **Split isolation:** the 100-case internal test is separated before model selection and remains untouched until the final assessment, i.e. with frozen configuration.
- **Fold-local preprocessing:** scalers and other preprocessing operations are fitted exclusively on the training portion of each fold natively by scikit-learn and custom implemented by functions built on top of PyTorch.
- **Strict OOF predictions:** every OOF prediction is produced by a model that was not trained on the corresponding example.
- **Training-only optimization:** hyperparameters and ensemble weights are selected exclusively using development-set cross-validation and OOF predictions.
- **Frozen assessment:** configurations and ensemble weights are fixed before the internal test is evaluated.
- **Blind-test isolation:** the 1,000 blind examples are never used for preprocessing, model selection, weight optimization or performance estimation.
- **Bootstrap uncertainty:** bootstrap resampling is applied only after all model choices have been frozen, providing an estimate of the variance and confidence interval of the final internal-test MEE without influencing model selection.

CUP model-selection — search space definition

All searches use only the 400-case development set. Scikit-learn configurations are selected by five-fold GridSearchCV while nested 5×3-fold CV is used as a less biased performance check. The NN search is organized as sequential Optuna studies eventually verified on the cup_NN_manual.ipynb with a combination of configuration – seed to explore metric consistency.

Model	Search and validation	Explored hyperparameters	Evaluations and criterion
KNN	5-fold GridSearchCV; nested 5×3 check	k=1–39 odd; p={1,2}; uniform; Standard/Robust scaling	80 configurations; min. mean CV MEE
SVR	5-fold GridSearchCV; nested 5×3 check	Custom RBF/RBF/poly.; $C=10^{-2}$–10^{4}; $\gamma=10^{-4}$–3/scale; ε=.01–.8; degree 2–3	348 configurations; min. mean CV MEE
LinearSVR	5-fold GridSearchCV; nested 5×3 check	$C=10^{-4}$–10^{3}; ε-insensitive/squared; dual={T,F}; 20k iterations	24 configurations; min. mean CV MEE
NN	Sequential Optuna; 3/5-fold CV; MedianPruner	η and λ: coarse→local; batch=16–128; 13 architectures;Activations: ReLU/Tanh/GELU	160 trials; VL MEE + TR–VL gap diagnostic

CUP model-selection — Optuna sensitivity to training seed

With fixed Optuna sampling and CV folds, predictive performance remains relatively stable while the selected NN configuration changes across training seeds. This suggests that there are several regions of search space with comparable performance. The three finalists were therefore be re-evaluated under every seed on the dedicated cup_NN_manual.ipynb.

Run / training seed	MEE (TR / CV) · gap	Architecture	Optimization
Run 1 · seed 1	16.0696 / 21.0251 Gap 4.9554	[256] · Tanh	Batch 64 · LR 0.10855 WD 4.67×10^{-5}
Run 2 · seed 10	13.9237 / 20.3317 Gap 6.4079	[128, 64] · GELU	Batch 16 · LR 0.04678 WD 6.55×10^{-3}
Run 3 · seed 20	17.4577 / 21.4611 Gap 4.0034	[64, 64] · GELU	Batch 32 · LR 0.10890 WD 5.27×10^{-6}
Across runs	CV MEE 20.939 ± 0.465 Mean gap 5.122	Three distinct optima	Stable error range; seed-sensitive choices

CUP model-selection — NN frozen-model assessment

- **Performance evidence**
  - The frozen [128, 64] GELU network reduced MEE from the 35.44 baseline to 22.83.
  - Cross-validation and the diagnostic hold-out agreed closely: 20.28 ± 0.82 versus 20.08 MEE.
  - The bootstrap OOB estimate, 22.72 ± 1.09 MEE, almost matched the independent internal result ($\Delta = 0.11$).
  - The internal result was 2.55 MEE worse than CV, revealing moderate CV optimism on the 100 held-out cases.
- **Interpretation and decision**
  - Final training reached the CV-derived target at epoch 171 (TR MEE = 15.52). The 7.31 TR–internal gap indicates residual overfitting, but bootstrap agreement and the strong baseline improvement support useful generalization. Configuration, training procedure and seed were frozen before assessment; the internal result is not used for further tuning.
  - There was insufficient evidence to override the systematically selected Optuna configuration. Any manual configuration we tried seemed to break the trade off between the hyperparams combination. Suggesting what we well know: the hyperparams are strictly interconnected.

CUP model-selection — results and hyperparameters

On the untouched 100-case internal test, KNN remains the strongest individual model. RBF-SVR stays competitive, while the NN shows the largest generalization error. LinearSVR was eventually excluded form ensemble for poor performance and used more as an intermediate baseline.

Model	Architecture and selected hyperparameters	MEE (TR / internal TS)	CV validation MEE
NN	128–64–32 GELU; AdamW; $\eta=0.008375$; $\lambda=1.05\times10^{-5}$; mini-batch 16	12.7045 / 22.0251	20.0802 ± 0.4017
KNN	k=5; Euclidean (p=2); uniform weights; StandardScaler	13.2870 / 15.4430	17.1390 ± 0.2021
RBF-SVR	C=100; $\gamma=3$; $\varepsilon=0.1$; four independent outputs	0.2599 / 18.1632	17.6784 ± 0.6903
LinearSVR*	C=0.01; squared ε-insensitive; dual=False; max_iter=20,000	24.9427 / 26.8403	25.1892 ± 0.8073

CUP model selection — complementary diagnostics

Complementary OOF metrics broadly preserve the same evidence: KNN and RBF-SVR are closely matched, while NN shows larger residuals. R^2 was not reported in the latest NN run.

Diagnostic	KNN	RBF-SVR	NN
MAE	7.2924 — lowest; supports best MEE.	7.7661 — slightly higher; supports MEE ranking.	8.8193 — highest; supports worst MEE.
RMSE	10.0911 — lowest; limited large residuals.	10.1046 — near-tie with KNN.	11.5288 — higher; larger residuals.
MSE	102.2469 — lowest; near-tie with SVR.	102.2488 — equivalent; MEE differentiates.	134.4837 — much higher; supports weaker MEE.
R^2	0.6538 — ≈65% variance explained.	0.6559 — negligible edge; no contradiction.	Not reported in latest NN run.

CUP bootstrap stability — frozen individual models

Bootstrap OOB resampling (100 replicates) was performed on the development set after model selection. Its mean and dispersion complement—not replace—the single frozen internal-test assessment. Comparable estimates support stability; differences reflect resampling and finite-sample variability.

Model	Internal-test MEE	Bootstrap OOB MEE (mean ± SD)	Relative SD
KNN	16.8191	17.66 ± 1.09	6.16%
RBF-SVR	18.1632	18.45 ± 0.75	4.07%
NN	22.8320	22.72 ± 1.09	4.78%
LinearSVR	26.8403	25.22 ± 0.73	2.91%

CUP Ensemble — Final MEE summary

Training MEE is in-sample and optimistic; the untouched internal test is the primary generalization estimate. The ensemble preserves its OOF improvement on the internal split.

Model	Training	OOF validation	Internal test
KNN	10.5313	16.3223	16.8191
RBF-SVR	2.3408	17.7843	17.8229
NN	15.5079	20.2779	21.7458
Ensemble	**9.0248**	**16.0993**	**16.5450**

Final model choice — frozen weighted ensemble

Final CUP predictor: weighted ensemble of KNN, RBF-SVR and NN.

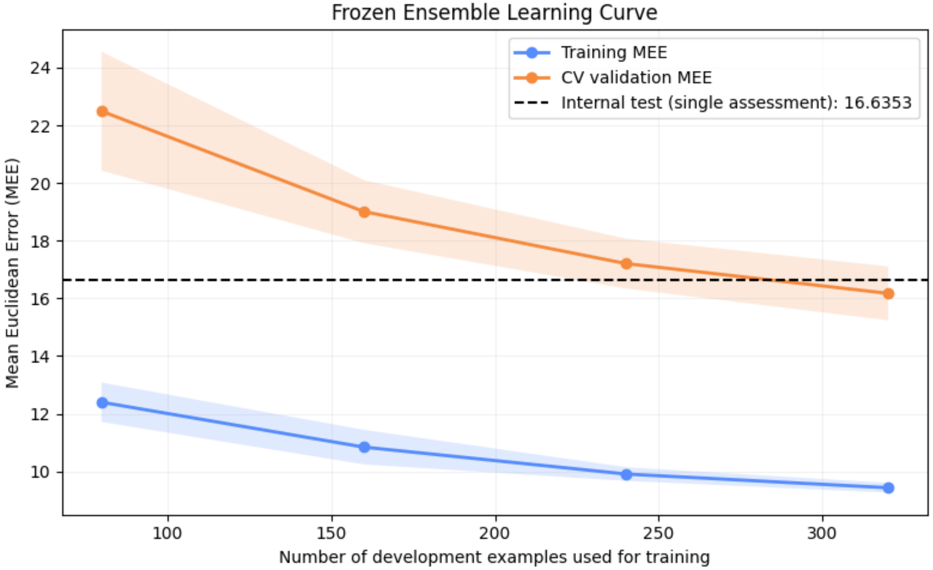
- Evidence: OOF MEE 16.0748 versus 16.3223 for KNN; internal-test MEE 16.6436 versus 16.8191.
- Rationale: KNN is the strongest standalone model, while RBF-SVR and NN add complementary residual information.
- Exclusion: LinearSVR was removed because of consistently weaker performance and considered as a model baseline.
- Frozen OOF weights: KNN = 73.84%, RBF-SVR = 16.06%, and NN = 10.10%; the non-negative weights are constrained to sum to 100%.
- Leakage control: components and OOF-derived weights were frozen before the single internal-test assessment and blind-test prediction.

Final training: after selection, every component was refitted on all 500 labelled examples and combined using the frozen weights.

Decision: the ensemble was selected because it delivered a consistent, independently confirmed improvement over the strongest standalone baseline.

Ensemble Learning Curve

Since the final predictor is a weighted ensemble without an epoch-based training process, performance is analysed as a function of the training-set size. The persistent training–validation gap indicates some overfitting. Nevertheless, CV validation MEE steadily decreases and its variability narrows as more training examples become available suggesting that with a larger training set the gap might be reduced.



Discussion — findings and limitations

- Weight initialization: the custom init_weights function was used to compare different initialization schemes. We forced initializing all weights to zero to experiment degenerated results in and unstable training behaviour. The final neural networks used Kaiming-uniform weight initialization, while biases were initialized to zero (see appendix for a concrete example).
- Generalization gap: training MEE remains below CV validation MEE, indicating residual overfitting.
- Learning curve: validation MEE falls as the training set grows, suggesting that additional labelled data could improve generalization.
- Ensemble gain: improvement over KNN is modest—about 1.5% OOF and 1.0% on the internal test—but consistent.
- Complementarity: lower-weight RBF-SVR and NN predictions help because their residuals differ from those of KNN.
- Uncertainty: the internal test is a single held-out assessment, not a variance-free performance estimate.

Main limitations: limited data and a narrow performance margin. The ensemble provides a reproducible incremental improvement.

Appendix — Uniform Weight Initialization

Uniform zero-weight (or constant-weight) initialization causes the network to learn nearly identical representations. Below the learning curve and the gradient norm resulting from a run on MONK-1 with all weight set to zero.